

DSA INTERVIEW MASTER PLAN

Pattern Recognition Strategy

Focus on recognizing patterns instead of memorizing solutions.

Starter Pattern Problems:

- Two Sum
- Maximum Subarray (Kadane's Algorithm)
- Binary Search
- Valid Parentheses
- Merge Two Sorted Lists

WEEK 1: Arrays & Basic Hashing

Key Topics

- Prefix Sum
- Two Pointer Technique
- Sliding Window
- HashMap-based problems

Must-Solve Problems

- Two Sum (LeetCode #1)
- Best Time to Buy and Sell Stock (LeetCode #121)
- Maximum Subarray (LeetCode #53)
- Subarray Sum Equals K (LeetCode #560)

WEEK 2: Sorting & Searching

Key Topics

- Binary Search on Answer
- Custom Sorting
- Search Space Problems

Must-Solve Problems

- Binary Search (LeetCode #704)
- Find First and Last Position (LeetCode #34)
- Search in Rotated Sorted Array (LeetCode #33)
- Kth Largest Element (LeetCode #215)
- Merge Sorted Arrays (LeetCode #88)
- Find Peak Element (LeetCode #162)

WEEK 3: Strings

Key Topics

- Sliding Window
- HashMap Counting
- Palindrome & Anagram

Must-Solve Problems

- Valid Anagram (LeetCode #242)
- Longest Substring Without Repeating (LeetCode #3)
- Group Anagrams (LeetCode #49)
- Minimum Window Substring (LeetCode #76)

WEEK 4: Linked List

Key Topics

- Fast & Slow Pointers
- Reversal
- Cycle Detection
- Merging

Must-Solve Problems

- Reverse Linked List (LeetCode #206)
- Detect Cycle (LeetCode #141)
- Merge Two Sorted Lists (LeetCode #21)
- Remove Nth Node (LeetCode #19)

WEEK 5: Stack & Queue

Key Topics

- Monotonic Stack
- Next Greater Element
- Stack Simulation

Must-Solve Problems

- Valid Parentheses (LeetCode #20)
- Min Stack (LeetCode #155)
- Daily Temperatures (LeetCode #739)
- Largest Rectangle in Histogram (LeetCode #84)

WEEK 6: Trees (Binary & BST)

Key Topics

- Traversals
- Height & Diameter
- LCA
- BFS / DFS

Must-Solve Problems

- Maximum Depth (LeetCode #104)
- Diameter of Tree (LeetCode #543)
- Symmetric Tree (LeetCode #101)
- Validate BST (LeetCode #98)